

Bulk solids, surfaces and vessel constraints

Read and prepare: 15–20 minutes total. Bring tentative answers; uncertainty is useful for discussion.

What fixes the gas pressure when a bulk solid can appear or disappear?

Three different gas-solid questions

Bulk solid-vapor equilibrium transfers the same substance between phases, as in sublimation. Adsorption describes uptake at a surface. A gas-solid reaction changes chemical species and needs reaction stoichiometry and an equilibrium constant. Name the process before choosing equations. Lab 10 treats nonreacting bulk equilibrium; Lab 11 treats adsorption. The current reaction-extent solver does not solve general heterogeneous reactions.

The vessel matters

For a rigid closed vessel at fixed T and V , equilibrium minimizes Helmholtz energy under the material constraints. For fixed T and P , the corresponding potential is Gibbs energy. In the ideal-gas sublimation model, solid present at equilibrium fixes the subliming species partial pressure at $P_{\text{sub}}(T)$. There must be enough total material to fill the gas at that pressure.

Depletion and other equilibria

If the available amount is below the saturated gas capacity, the solid disappears and the gas is undersaturated. Inert gas contributes to total pressure in this ideal model. For SLE, identify the solid standard state and liquid activity. A bulk solubility condition and an adsorption loading equation describe different phase inventories.

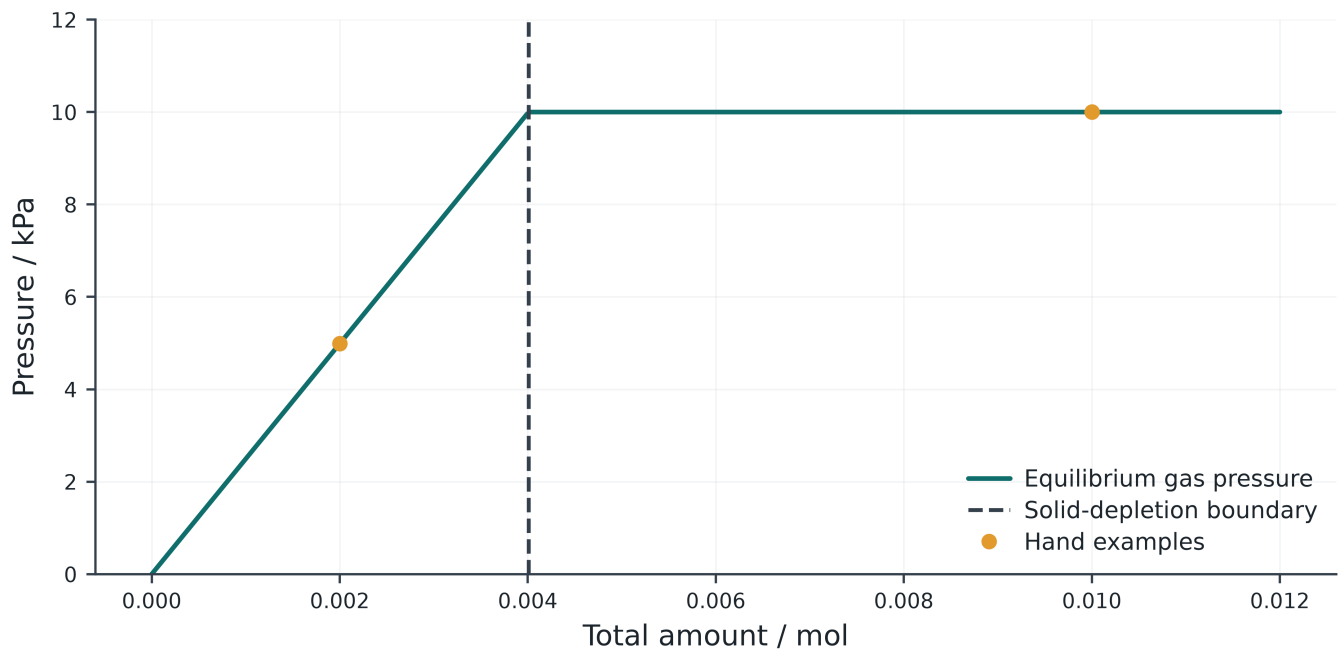
$$n_{\text{g,sat}} = P_{\text{sub}} V / (RT); n_{\text{g}} + n_{\text{s}} = n_{\text{total}}$$

$$P = n_{\text{total}} RT / V \text{ when the solid is exhausted (no inert gas)}$$

One example to reason through

Synthetic teaching values. These numbers illustrate the assumptions; they are not measurements of a real material.

Assume a synthetic material at 300 K with $P_{\text{sub}}=10.0$ kPa in a 1.00 L gas volume. Use $R=8.314462618$ kPa L mol⁻¹ K⁻¹. Saturated gas requires 0.00400908 mol. With 0.0100 mol total, the gas remains at 10.0 kPa and 0.00599092 mol is solid. With only 0.0020 mol, all material is gas and $P=4.98868$ kPa.



Check and interpret

For a separate Langmuir teaching example, $q_m=2.00$ mmol/g and $b=0.100$ kPa⁻¹ give $q=q_m bP/(1+bP)=1.00$ mmol/g at 10 kPa. To close a vessel balance, multiply loading by adsorbent mass and convert mmol to mol. This surface amount is not the bulk solid remaining in the sublimation calculation.

Bring three short answers

A sketch, a few numbers or one sentence is enough. Do this before opening the calculator; revise your prediction in class.

1. Classify sublimation, adsorption and a gas-solid reaction by what changes physically.

2. Predict pressure when the total amount increases from 0.002 to 0.010 mol in the example.

3. Which thermodynamic potential matches a rigid closed vessel at fixed temperature?

Optional extension

Henry gas-solubility exercise: using the explicitly defined convention $p=H \cdot x$ with $H=1000$ kPa, find x for $p=10$ kPa. Distinguish this H from a Henry adsorption coefficient. Osmosis exercise: for an ideal dilute nonelectrolyte, estimate $\Pi=cRT$ at $c=10$ mol/m³ and $T=300$ K. These are paper extensions, not implemented new lab modes.

Continue in the core lab

<https://www.skhgroup.net/teaching/2105603/labs/solids.html>

Reading locator: supplied textbook contents, sections 16.1, 16.5, 16.8-16.9; compare 17.5.

Original course explanation and examples, not a reproduction of the textbook. Section numbers are based on the supplied contents pages. Required weekly independent work: 15–20 minutes preparation plus 30–40 minutes follow-up; optional extensions are additional.